Summary in Graph

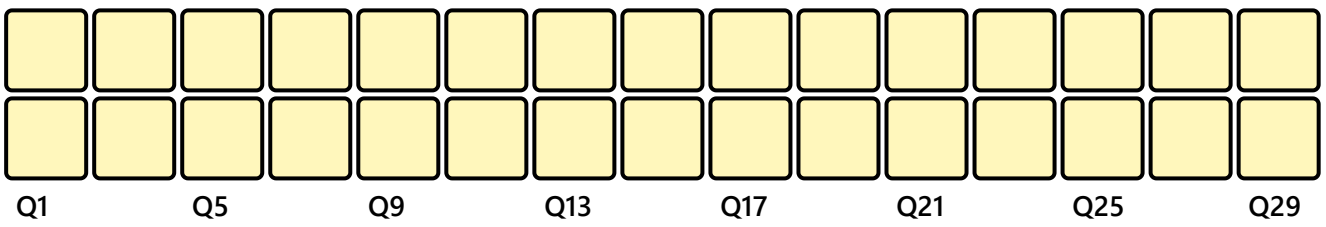
Exam Summary (GO Classes CS Test Series 2026 | Algorithms | Subject Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (20) out of 196 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Algorithms

```
int func(int n) {
    if (n == 0) {
        return 0;
        }
5.  else {
            return n % 2 + func(n / 2);
        }
}
```

Consider the following C program.

What will be the output of func(128)?

A. 0

B. 1

C. 2

D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Let $A1,\ A2,\ A3$, and $A4$ be the five matrices of dimensions $10 \times 5, 5 \times 20$, 20×10, and 10×20 respectively. Using the standard matrix multiplication approach and choosing the optimal parenthesization, the minimum number of scalar multiplications required to compute this product is _____ ?

Your Answer: | Correct Answer: 3000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

A System of difference constraints has inequalities of the form $x_j - x_i <= C$. These can be represented using a directed graph $G = (V, E)$, where each variable is a node and each constraint becomes an edge x_j to x_i with weight C.

Consider the following system of constraints:

$x_1 - x_2 <= 2$

$x_2 - x_3 <= 1$

$x_3 - x_1 <= -3$

$x_1 - x_4 <= 4$

What can you conclude about the system?

A. The system has no feasible solution because of the negative cycle.

B. The system has unique feasible solution

C. The system has infinitely many feasible solutions.

D. The system has feasible solutions only if $x_1 = x_2 = x_3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following statements about the Depth-first search is/are TRUE?

- **S1**. The running time of depth-first search in a complete graph, as a function of $|V|$ and $|E|$, if the input graph is represented by an adjacency list, is $O(|V|^2)$
- **S2**. If the depth-first search tree of an undirected graph $G(V, E)$ has exactly one back edge and no other edges then the G has exactly one cycle.

A. Both S1 and S2 are True

B. S1 is True and S2 is False

C. S1 is False and S2 is True

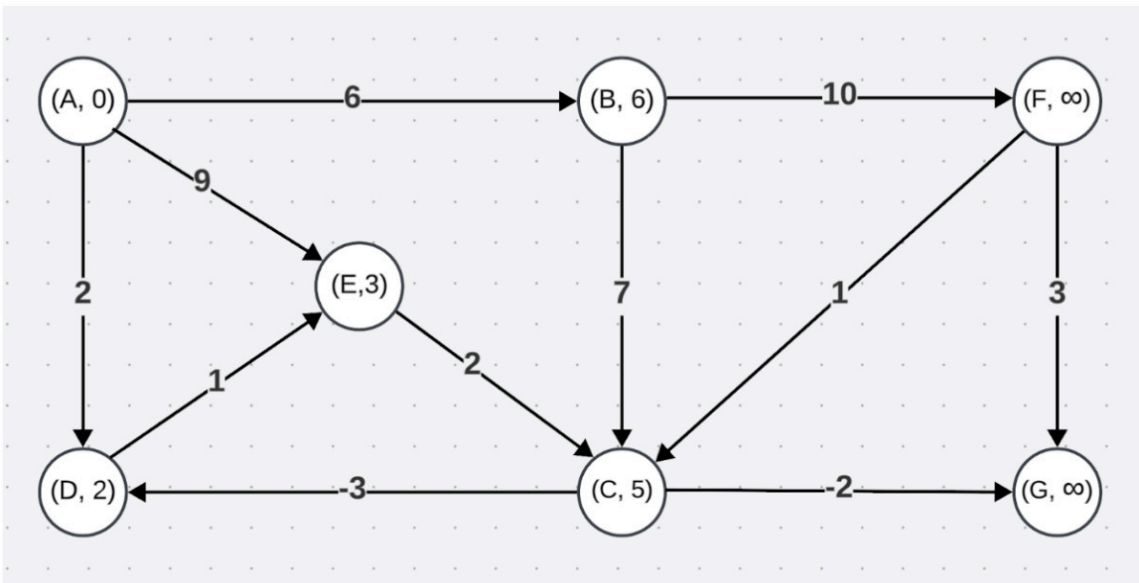
D. Both S1 and S2 are False

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the directed weighted graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ shown below. Each vertex is represented as (X, d), where X is the node label and d denotes the current shortest distance estimate from the source vertex (A, 0).

At an intermediate stage of Dijkstra's algorithm, after extracting k vertices from the priority queue, the vertex $(\mathrm{k}+1)$ th is extracted. What will be the updated distance estimates of all vertices immediately after relaxing all outgoing edges from the $(\mathrm{k}+1$)th vertex?



A. $(A, 0), (B, 6), (C, 5), (D, 1), (E, 3), (F, \infty), (G, 0)$

B. $(A, 0), (B, 6), (C, 5), (D, 1), (E, 3), (F, \infty), (G, 3)$

C. $(A, 0), (B, 6), (C, 5), (D, 2), (E, 3), (F, \infty), (G, 3)$

D. $(A, 0), (B, 6), (C, 5), (D, 2), (E, 3), (F, \infty), (G, 0)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following two statements:

- **S1**: If root vertex r has more than one child of a DFS(Depth First Search) tree of an undirected graph G , then removing r will disconnect the graph.
- **S2**: In a BFS(Breadth First Search) tree of a connected undirected graph G, for any vertex v, all its neighbours are at depth at most one.

Which ONE of the following is CORRECT?

A. Both S1 and S2 are TRUE

B. S1 is TRUE and S2 is FALSE

C. S1 is FALSE and S2 is TRUE

D. Both S1 and S2 are FALSE

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

$\mathrm{f(n)}$ is $\mathrm{O(g(n))}$ if there exists some values c , $n_o > 0$ such that $\mathrm{f(n)} \leq \mathrm{c(g(n))}$ for all $\mathrm{n} \geq n_o$.
Which of the following are True about this definition? Select all that apply.

A. c and n_0 cannot have the same value

B. c and n_o must be integers.

C. If $\mathrm{f(n)} = \mathrm{g(n)}$, then $\mathrm{f(n)} \in \mathrm{O(g(n))}$

D. $\mathrm{g(n)}$ cannot be $\mathrm{O(1)}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE?

- **S1**: The running time of a dynamic programming algorithm is always $\Theta(\mathrm{P})$ where P is the number of subproblems.
- **S2**: For a dynamic programming algorithm, computing all values in a bottom-up fashion is asymptotically faster than using recursion and memoization.

A. S1 is True and S2 is False

B. S1 is False and S2 is True

C. Both S1 and S2 are True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider an unordered array of N distinct integers. What is the minimum number of comparisons required to identify all elements except the largest?

A. N - 1

B. N - 1 + $\lceil \log_2 N \rceil$

C. N - 1 + $\lfloor \log_2 N \rfloor$

D. N + $\lceil \log_2 N \rceil - 2$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following statements for undirected weighted graph G :

- **S1**: Each edge of G has weight at least 1. If there are exactly two edges with weight 1 then any minimum spanning tree contains both the edges.
- **S2**: Edge $\mathrm{e} = (\mathrm{u}, \mathrm{v})$ does not belong to an MST of G if and only if u and v can be joined by a path consisting entirely of edges that are cheaper than e.

Which of the statements are TRUE.

A. S1 is True and S2 is False

B. S1 is False and S2 is True

C. Both S1 and S2 is True

D. Both S1 and S2 is False

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 05sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the code fragment written in C below

```
void fun(int n){
    if (n == 0){
        printf("0");
    }
    if( n == 1){
        printf("1");
    }
    else{
        fun(n/2);
        printf("%d", n%2);
    }
}
```

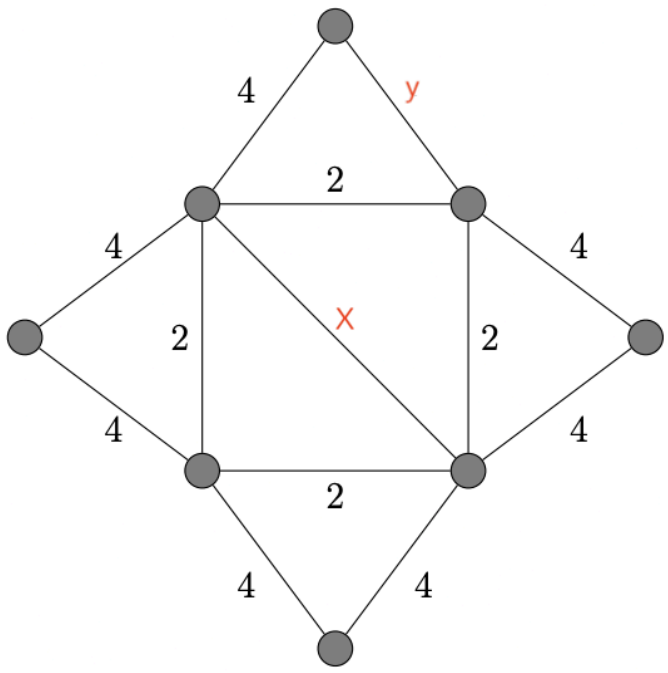
What does fun(123) print?

A. 1111001
B. 1111111
C. 1111101
D. 1111011

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let the minimum spanning tree (MST) of the graph shown below contain the edge weights x and y.



What will be the possible range of x and y if all edge weights are non-negative?

A. $0 <= \mathrm{x} <= 2$ and $0 <= \mathrm{y} <= 4$

B. $1 <= x < 2$ and $1 <= y < 4$

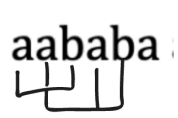
C. $1 <= \mathrm{x} <= 2$ and $1 <= \mathrm{y} <= 4$

D. $0 <= \mathrm{x} < 2$ and $0 <= \mathrm{y} < 4$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let $\mathrm{C(S, i, j)}$ be the function that counts the number of subsequence w , string S have and i, j are the pointers pointing to the last character of the string S and W respectively.
For example, if $S = aababa$ and $w = aab$ then there are 2 subsequences of w are present



Which recurrence relation correctly computes C(S, i, j)?

A.

$$C(S,i,j) = \begin{cases} 0 & \text{if i} = 0\&\text{j} \neq 0 \\ 1 & \text{else if } j = 0 \\ C(S,i-1,j-1) + C(S,i-1,j) & \text{else if } s[i] = w[j] \end{cases}$$

B.

$$C(S,i,j) = \begin{cases} 0 & \text{if i} = 0\&\text{j} \neq 0 \\ 1 & \text{else if } j = 0 \\ C(S,i-1,j-1) + C(S,i,j) & \text{else if } s[i] = w[j] \end{cases}$$

C.

$$C(S,i,j) = \begin{cases} 0 & \text{if i} = 0\&\text{j} \neq 0 \\ 1 & \text{else if } j = 0 \\ 1 + C(S,i-1,j-1) + C(S,i,j) & \text{else if } s[i] = w[j] \end{cases}$$

D.

$$C(S,i,j) = \begin{cases} 0 & \text{if i} = 0\&\text{j} \neq 0 \\ 1 & \text{else if } j = 0 \\ C(S,i-1,j) + C(S,i,j-1) & \text{else if } s[i] = w[j] \end{cases}$$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the recursive quicksort algorithm with "random pivoting". That is, in each recursive call, a pivot is chosen uniformly at random from the sub-array being sorted. When this randomized algorithm is applied to an array of size "n" all whose elements are distinct, what is the probability that the smallest and the largest elements in the array are compared during a run of the algorithm?

A. $\frac{1}{n}$

B. $\frac{2}{n}$

C. $\theta(\frac{1}{n.logn})$

D. $\theta(\frac{1}{n^2})$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15

Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements are TRUE.

A. While sorting an array A of n elements using merge sort, the number of element comparisons at each level of the merge tree is $\Theta(\log n)$ in the worst case.

B. In worst-case quicksort, the number of comparisons is equal to $\Theta\left(n^2\right)$, Which cause the recursion tree to be grow to a depth of $n-1$

C. The average number of swaps in selection sort is $\Theta(n)$ regardless of input size.

D. Heapsort can be implemented to be stable and in-place simultaneously.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

An array A of n real distinct numbers is said to be quadratically true if the elements of the array represent values of some quadratic function, i.e.,

$$A[i] = ai^2 + bi + c \quad \text{for constants } a, b, c \text{ and } 0 \leq i < n$$

Assume that $\mathrm{a}, \mathrm{c} > 0$ and $\mathrm{b} < 0$, the function represents a convex parabola, and the array has a single local minimum.
What is the best possible asymptotic bound to find the index of the local minimum in such an array?

A. $\theta(logn)$

B. $\theta(n)$

C. $\theta(1)$

D. $\theta\left(\log^2 \mathrm{n}\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let A be a one-dimensional array of n distinct real numbers such that $A[i] = f(i)$, where $f : \mathbb{R} \to \mathbb{R}$ is an arbitrary but well-behaved function. It is known that:

- $f(x)$ is continuous and differentiable in the closed interval $[a, b]$,
- $f(x)$ has exactly two real roots in $[a, b]$,
- There exists a unique point $p \in [a, b]$ such that:

$$\lim_{h \to 0^+} f'(p+h) < 0, \quad \lim_{h \to 0^-} f'(p-h) > 0, \quad \text{and } f'(p) = 0$$

Given that the array A is indexed over integers in $[a, b] \cap \mathbb{Z}$, what is the minimum time complexity (in terms of n) required to find the index i closest to the point p where $f(x)$ achieves its maximum value?

A. $\theta(logn)$

B. $\theta(n)$

C. $\theta(1)$

D. $\theta(log^2(n))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 2 | Penalty: 0 | Algorithms

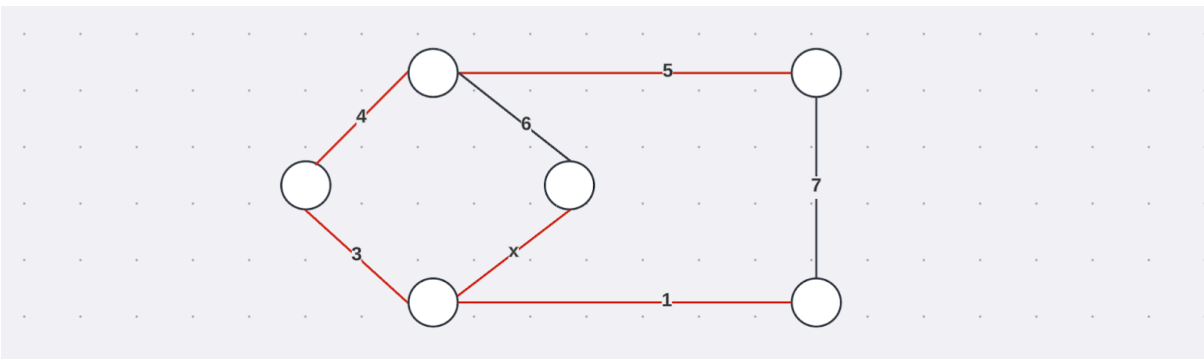
Below is the frequency-encoding table for a Huffman Tree T. The table has a missing entry for character ***I*** in the frequency field. How many possible values the missing entry can take?

char	freq	encoding
B	2	01111
F	1	01110
H	3	0110
I	?	00
L	5	010
M	15	10
S	15	11

Your Answer: | Correct Answer: 10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

For the following positive weighted graph G the red line forms the minimum spanning tree.



which of the following is True about the range of x?

A. $[1, 7]$

B. $[4, 6]$

C. $[1, 6]$

D. $[5, 6]$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 2 | Penalty: 0 | Algorithms

Consider a simple undirected connected bipartite graph $\mathrm{G} = (\mathrm{V}, \mathrm{E})$ with $|\mathrm{V}| = 6$ (3 vertices in each partition) and $|\mathrm{E}| = 9$, where all edge weights are distinct. What is the maximum possible rank of an edge (i.e., its position in the sorted list of edge weights) that can be included in the Minimum Spanning Tree (MST) of G?

Your Answer: | Correct Answer: 7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements are TRUE for an undirected weighted graph G?

A. If G has exactly two edges with the same weight, then it must have more than one MST.

B. Every edge of the minimum spanning tree is either a cut edge or a unique lightest edge.

C. If an edge $e \in E$ is the maximum-weight edge in some cycle, then e does not belong to any Minimum Spanning Tree of G.

D. The complement of a Minimum Spanning Tree $T \subseteq G$ (i.e., $E \backslash T$) contains only edges that form cycles with the edges in T.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $G = (V, E, w)$ be a connected undirected graph with distinct positive integer weights and Gk be the subgraph formed by removing edges with weight k or larger. Define k as the maximum integer k such that Gk is disconnected. Then, which of the following is always true?

A. $k = \max e \in MST(w(e) + 1)$

B. $k = \max e \in MST\ w(e)$

C. $k = \max e \in E$
$w(e)$

D. $k =$ any edge weight from set $\{$all greater than max
$e \in MST$
$w(e)\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are FALSE?

A. If all edge weights of $G = (V, E)$ are distinct, the MST of a graph G is unique.

B. Removing the maximum-weight edge in MST always disconnects the graph $G = (V, E)$.

C. For a weighted connected undirected graph $G = (V, E)$, if any subgraph Gk (formed by removing edge weight k and larger) is disconnected, then MST of G always contains only edge weights $<= k$.

D. In any MST of a connected graph $G = (V, E)$, the number of edges is always exactly $|V| - 1$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $A = [a0, a1, a2, \ldots, an]$ be an unsorted array where n is a power of 2. Suppose Quick Sort is applied to this array, and at every recursive step, the partition function always divides the array in 1:n. What will be the time complexity of the Quick Sort implementation?

A. $\Theta(n)$

B. $\Theta(nlogn)$

C. $\Theta\left(n^2\right)$

D. $\Theta(\log n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following statements is/are TRUE?

A. A DFS tree always has exactly $\mathrm{n}-1$ edges if the graph G has n vertices.

B. Let T be a depth -first search tree of an undirected graph G . Let (x, y) be an edge of G that is not an edge of T, then one of x or y is an ancestor of the other.

C. If an edge (u, v) in an undirected graph G is not a part of the DFS tree and neither u is an ancestor of v nor v is an ancestor of u in the DFS tree, then the edge (u, v) is a cross edge.

D. In a DFS if vertex u is discovered before vertex v then there is an edge (u, v) in G and either u or v are ancestors of others in the DFS tree.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Which of the following statements are TRUE?

- **S1**: Run Bellman-Ford on a weighted graph $\mathrm{G}=(\mathrm{V}, \mathrm{E}, \mathrm{w})$ from a vertex $\mathrm{s} \in \mathrm{V}$. If there is a witness $\mathrm{v} \in \mathrm{V}$, i.e., $\delta_{|\mathrm{V}|}(\mathrm{s}, \mathrm{v})<\delta_{|\mathrm{V}|-1}(\mathrm{s}, \mathrm{v})$, then v is on a negativeweight cycle of G .
- **S2**: If there is an algorithm to solve $0-1$ Knapsack in polynomial time, then there is also an algorithm to solve Subset Sum in polynomial time.

A. S1 is True and S2 is False

B. S1 is False and S2 is True

C. Both S1 and S2 are True

D. Both S1 and S2 are False

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider procedures 1 and 2 expressed in pseudocode as Follows:

begin procedure 1
while n > 1 do
for i = 1 to n do
printf("hello")
end for
n = floor(n/2)
end while
end procedure
	begin procedure 2
	begin If n > 1
	printf("hello")
	begin
	procedure 2(floor(n/2))
	end
	end If
	end procedure

Let $\mathrm{f1(n)}$ and $\mathrm{f2(n)}$ denote the number of times the statement "printf("hello")" executed in procedure 1 and procedure 2, respectively.
Which of the following statements is/are TRUE?

A. $f1 \in O(f2)$

B. $f1 \in \Omega(f2)$

C. $f2 \in O(f1)$

D. $f1 \in \theta(f2)$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

$T(n) = \frac{9}{4}T\left(\frac{2}{3}n\right) + n^2$ and $T(1) = \Theta(1)$, then $T(n)$ belongs to

A. $\Theta(n \log n)$

B. $\Theta\left(n^2\right)$

C. $\Theta\left(n^2 \log n\right)$

D. $\Theta\left(n^2(\log n)\text{^}2\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + T\left(\frac{n}{6}\right) + O(n)$$
$$T(n) = 5 \quad \forall n < 100$$

Which of the following is the correct aysmptotic growth of $T(n)$?

A. $T(n) = \Theta(n \log n)$

B. $T(n) = \Theta\left(n \log^2 n\right)$

C. $T(n) = \Theta\left(n^2\right)$

D. $T(n) = \Theta\left(n^2 \log n\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the Code fragment of code written in C:

```
Int bar(int n){
    if(n <= 10){
        return 0;
    }
    if( n < 100){
        Int r = 0;
        for(int i = 0; i < n; i + +){
            r++;
        }
        return r;
    }
    return bar(n/3) + bar(n-1);
}
```

What will be the recurrence relation for the running time complexity of function $\mathrm{bar}(\mathrm{n})$, where n is in power of 2 ?

A.

$$T(n) = \begin{cases} 0 & n \leq 10 \\ c_2 n & n < 100 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_3 & \text{otherwise} \end{cases}$$

B.

$$T(n) = \begin{cases} 0 & n \leq 10 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_2 n + c_3 & n < 100 \end{cases}$$

C.

$$T(n) = \begin{cases} 0 & n \leq 10 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_2 n + c_3 & \text{otherwise} \end{cases}$$

D.

$$T(n) = \begin{cases} 0 & n <= 10 \\ c_2 \log_3(n) & n < 100 \\ T\left(\frac{n}{3}\right) + T(n-1) + c_3 & \text{otherwise} \end{cases}$$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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